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# Triassic paleomagnetic results from the Huanan Block, SE China

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**Abstract**

Paleomagnetic and rock magnetic investigations were undertaken on samples from 14 formations in an attempt to examine the tectonic history of south China and to evaluate competing hypotheses for the amalgamation of the Huanan and Yangtze Blocks. The vast majority of the sampled sites were remagnetized; however, results from the Lower Triassic Xikou Formation appear to have escaped remagnetization. The sites pass a fold test and are in good agreement with other results published by Gilder et al. [Gilder, S., Coe, R.S., Wu, H., Kuang, G., Zhao, X., Wu, Q., 1995. Triassic paleomagnetic data from south China and their bearing on the tectonic evolution of the western circum-Pacific region. Earth Planet. Sci. Lett. 131, 269–287]. From the data, however, it is difficult to constrain the hemisphere of these sites during the Early Triassic and as such, a firm resolution of the competing hypotheses cannot yet be offered. © 1999 Elsevier Science B.V. All rights reserved.

*Keywords:* Paleomagnetism; China; Tectonics; Huanan; Yangtze

## 1. Introduction

With its highly varied geology and accretionary tectonic history, China has proven to be an interesting tectonic puzzle. Geologic and paleomagnetic studies conducted by western as well as Chinese scientists, have revealed several distinct tectonic blocks (e.g., McElhinney et al., 1981; Lin et al., 1985; Yang et al., 1986; Hsü et al., 1988, 1990; Lin and Fuller, 1990; Enkin et al., 1992). Some of the most recent changes in the interpretation of South China tectonics have occurred as a result of geologic investigations by Hsü et al. (1988) and Gilder et al. (1993, 1995). The geology of the South China Block was reinterpreted by Hsü et al. (1988) with a collision melange reportedly being found in the rocks of the Banxi group. This collision melange separates the South China Block into two smaller blocks—the Yangtze Block in the northwest and the Huanan Block in the southeast. Gilder et al. (1993, 1995)

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have presented an alternative terrane model based on sinistral shear in the region after the Triassic.

The Yangtze Block is composed of the Sichuan foreland basin and the Yangtze deformed belt. The deformed belt is made up of detached and thrusted sedimentary cover. Though deformation is thought to have begun in the early Mesozoic, it continued until early Cretaceous and folded strata as young as Lower Cretaceous are recognized on this block.

The Huanan Block (formerly the South China Fold Belt) is a deformed block lying to the southeast of the Yangtze Block. Large scale folds and thrusts are common throughout the block, as are granitic intrusions. Deformation of sediments and rocks as

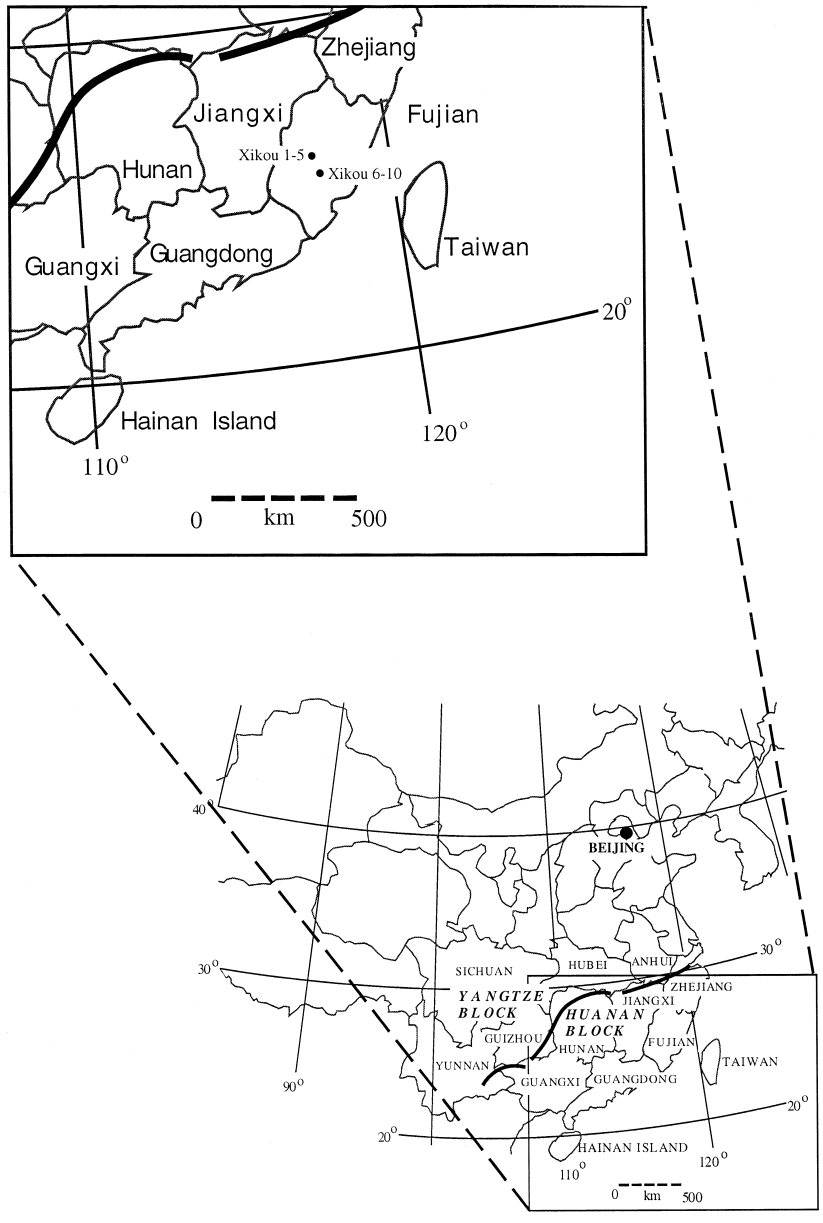



Fig. 1. Location and site map showing sampling sites for the Xikou Formation.

young as Lower Cretaceous is again seen and these deformed sediments are overlain by Cretaceous red beds.

At the contact between the Yangtze and Huanan Blocks, a collision melange has been discovered which contains metamorphosed rocks of the Banxi Group as well as ophiolites (Hsü et al., 1988). These rocks are thought by other geologists to be Precambrian basement (Anonymous, 1973; Gupta, 1989). Hsü et al. (1988) argued that at least some of what is called the Banxi Group belongs to a collision melange created during the collision of the two blocks in the early Mesozoic which obducted Precambrian oceanic crust.

The geologic evidence supporting an early Mesozoic collision and suturing of Huanan and Yangtze is disputed (Hsü et al., 1989; Gupta, 1989; Chen et al., 1991; Gilder et al., 1993, 1995). In order to test this new postulate, a paleomagnetic study of the Huanan Block was undertaken. Until now, there have been very few results reported from the block (Zhai et al., 1991, 1994; Dobson and Heller, 1992; Gilder et al., 1995).

## 2. Sampling and methodology

Samples from 14 formations and four provinces (Zhejiang, Fujian, Guangdong, and Hunan) were recovered for paleomagnetic analysis, however, results from only the Xikou Formation are presented here (Fig. 1). Most sites are located on the Huanan Block but the Yangtze Block was also sampled at several sites. The Yangtze Block results are discussed in a separate publication (Dobson and Heller, 1993; Dobson et al., 1993). Between five and 10 samples per site were drilled with a portable, water-cooled, gas-powered drill. The samples were oriented in place with an orientation device and a Brunton compass. All of the sites were taken from sedimentary sequences and most were limestones or siltstones; however, some sandstones were also sampled. Pilot samples were subjected to step-wise thermal demagnetization (usually from 13 to 18 demagnetization steps). In samples with narrow blocking temperature spectra, smaller increments were used.

The NRM of the samples was measured using an ScT and 2G Enterprises 3-axis SQUID magnetometer. After pilot studies, the remainder of the samples were demagnetized using between eight and 13 temperature steps depending on the blocking temperature spectra of the pilot samples. These data were plotted and analyzed using standard paleomagnetic techniques (Fisher, 1953; Zijderveld, 1967; Kirschvink, 1980).

Room-temperature Isothermal Remanence Acquisition (IRM) curves were generated by exposing representative samples from each site to incremen-

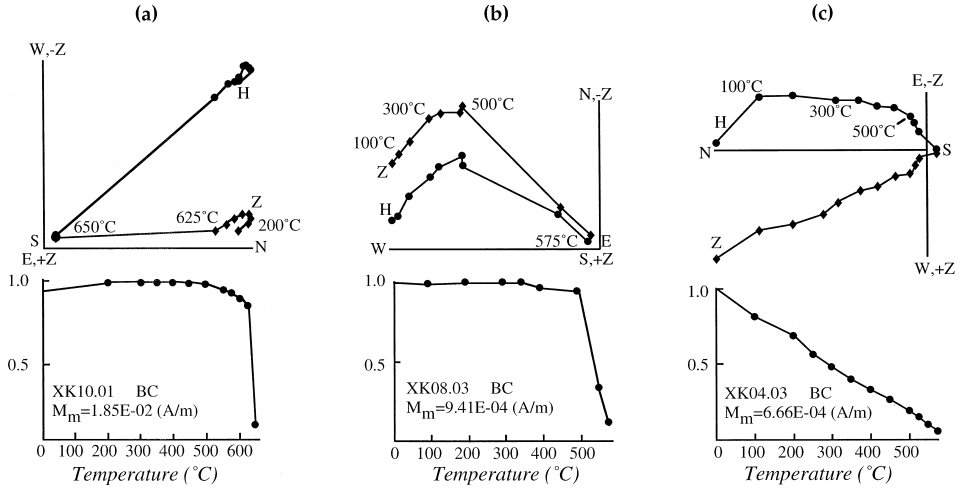



Fig. 2. Representative Zijderveld diagrams of thermal demagnetization for samples (a) XK10.01, (b) XK08.03 and (c) XK04.03 from the Xikou Formation.

tally increasing DC magnetic fields up to 1.0 T in an Oxford Instruments electromagnet. Afterwards, these samples were also subjected to step-wise thermal demagnetization in order to determine blocking temperatures.

## 3. Results

Of all of the formations studied, very few appear to accurately retain the geomagnetic field at the time of deposition. The majority of the sites have been remagnetized subsequent to deposition and this widespread remagnetization has been observed in nearly all studies of south China paleomagnetism (e.g., Lin et al., 1985; Opdyke et al., 1986; Heller et al., 1988; Steiner et al., 1989; Dobson and Heller, 1992; Gilder et al., 1995). It has been proposed that Mesozoic (Jurassic and post-Cretaceous) remagnetizations exhibiting single polarities found in two provinces in south China may be related to collision and suturing of the Huanan and Yangtze Blocks during that time (Dobson and Heller, 1992). A prominent Cretaceous overprint is thought to be a result of Yanshanian orogenic activity (Kent et al., 1987). A Brunhes overprint seen in many of the samples as well.

In this study, possible primary directions have been isolated in only one formation from the Huanan Block—the Lower Triassic Xikou Formation. The widespread remagnetization of the remaining formations sampled in this study, and its bearing on south China tectonics, is discussed in separate publications (Dobson and Heller, 1992; Dobson et al., 1993).

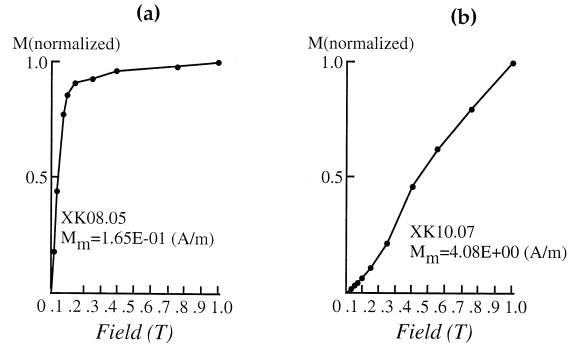



Fig. 4. Representative IRM acquisition curves for samples from the Xikou Formation showing samples with (a) low coercivity phases dominant and (b) high coercivity minerals dominant.

The Xikou Formation was sampled at 10 sites from two locations in Fujian Province near Yongan (sites 1–5; 25.8°N/117.5°E) and Qingliu (sites 6–10; 26.2°N/116.8°E). Most sites were taken in quarries and appeared quite fresh, though some sites were drilled from roadcuts. This formation consists of both siltstones and limestones, both of which were sampled. The bedding attitudes are considerably dispersed (from 224/19 to 15/87), which would enable a fold test to be performed.

During thermal demagnetization most samples exhibited multi-vectorial decay of NRM. In many cases a spurious component was removed by 100°C. Alternating field demagnetization, in general, was not as

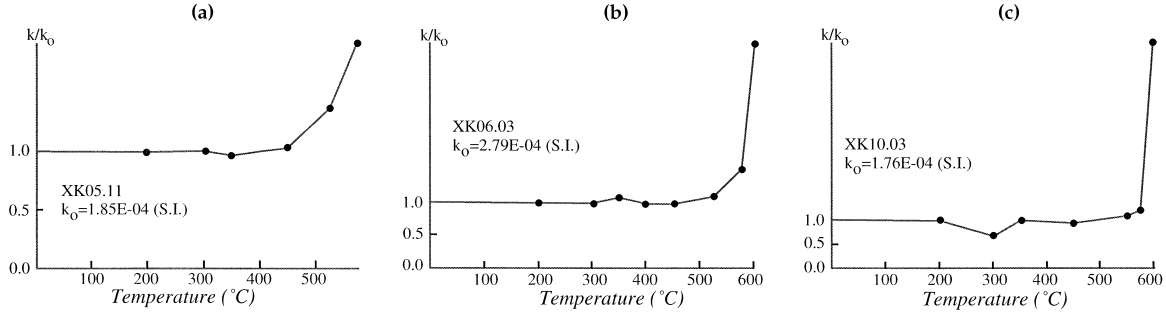



Fig. 3. Representative bulk susceptibility vs. temperature diagrams for samples (a) XK05.11, (b) XK06.03 and (c) XK10.03 from the Xikou Formation. $k_o$ is the initial susceptibility in S.I. units.

effective as thermal treatment for this formation and, therefore, was not used to a large extent. This is discussed in more detail in (Gilder et al., 1995). High temperature stable vectors were obtained for 50 samples from the formation. The temperatures at which these components were isolated varies but most laboratory unblocking temperatures exceeded 500°C and, in some cases, above 600°C (Fig. 2).

Bulk susceptibility was monitored in pilot samples from this formation during thermal demagneti-

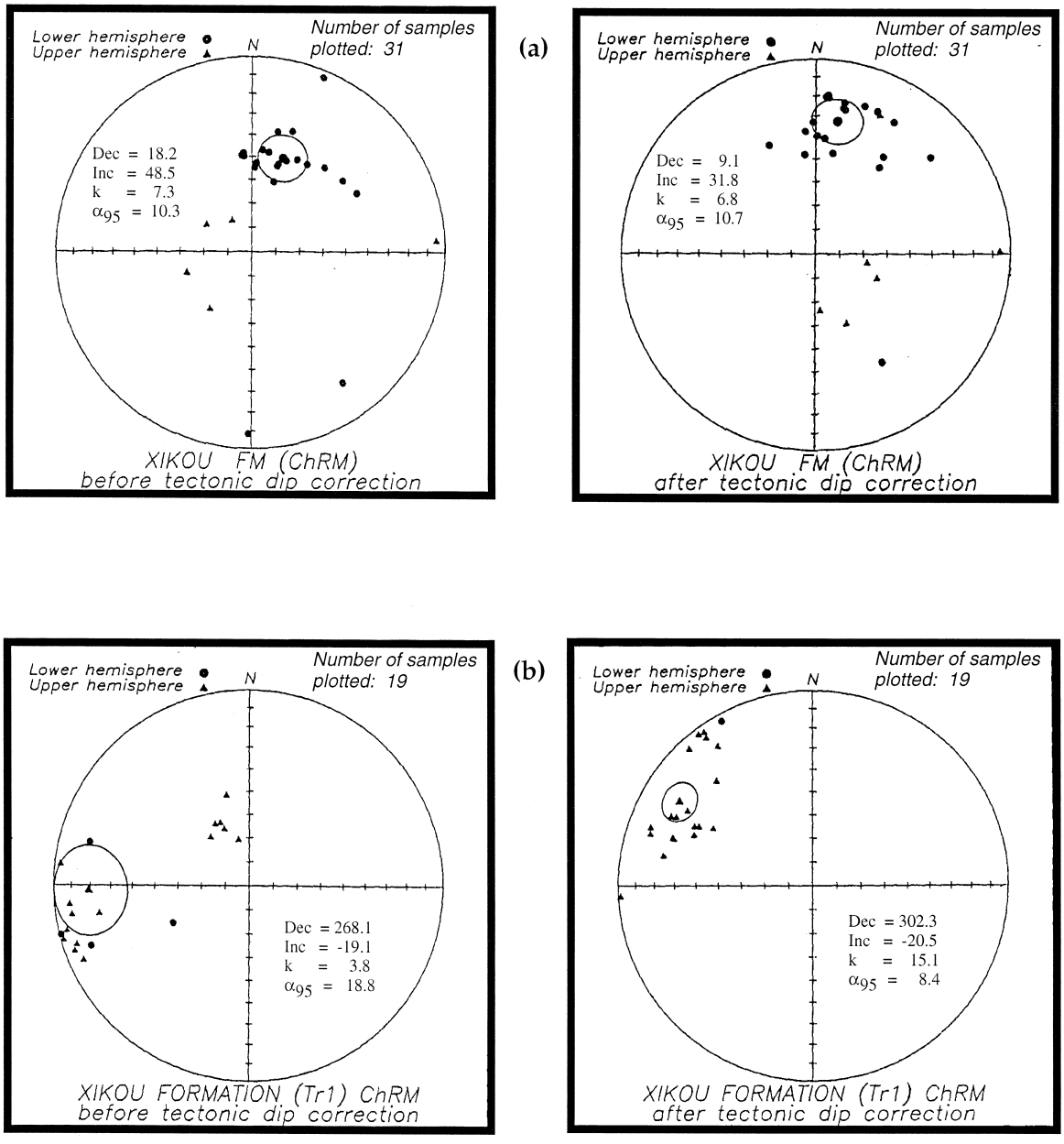



Fig. 5. Equal area projections of high temperature directions for (a) sites 1–5 and (b) sites 6–10 for the Xikou Formation.

zation. In most sites no increase was seen until temperatures above 600°C; however, in sites 1–5 an increase was seen above 450°C (Fig. 3). In some cases the increase in bulk susceptibility correlates with an increase in intensity during progressive thermal demagnetization making the isolation of the high temperature, stable component impossible for some samples. In cases where it was possible, the component isolated before the intensity increase was analyzed.

IRM acquisition curves show different behavior within this group of sites. Some are clearly dominated by low coercivity phases while others indicate the presence of high coercivity phases (Fig. 4). Thermal demagnetization of the IRM reveals both the presence of magnetite and hematite. Although some samples have unblocking temperatures between 525°C and 575°C, others become unblocked well above 600°C and, in some cases, near 700°C.

The high-temperature demagnetization vectors exhibit a large amount of scatter both before and after tectonic corrections (Fig. 5). Many of the samples have an in situ direction related to the present field while others are clearly not related. In view of the scatter seen in the directions and the fact that two geographic locations were sampled, these data were analyzed as two populations based on their location.

Sites 1–5 were sampled relatively close to each other near Yongan and the high-temperature decay vectors from these sites have been plotted together (Fig. 5a). The mean of the in situ directions lies near the present field direction, though there is considerable scatter in the individual directions.

An analysis of the high-temperature components from sites 6–10 (near Qingliu) shows clearly that these sites do not carry a recent overprint, as the in situ directions differ from the present axial dipole direction (Fig. 5b). Furthermore, a four-fold increase in the $k$ value upon correction for bedding attitude indicates that the remanence appears to have been acquired before folding at these sites. The bedding corrected declination for these sites is 302.3° with an inclination of $-20.5°$ and an $\alpha_{95}$ of 10.7.

## 4. Discussion and conclusions

Five sites from the Lower Triassic Xikou Formation pass a fold test at the 95% confidence level. Another result reported for this formation has been used as evidence of separation of the Huanan and Yangtze Blocks but yields a shallower paleolatitude and with a rotation in the opposite sense (Zhai et al., 1991, 1994). Details of this study, however, have not been published and the problems associated with it are discussed in (Gilder et al., 1995). Recently, Gilder et al. (1995) reported results from the Xikou Formation which are very similar to ours and are reported to pass the fold test; however, as with this study, only one polarity is present. Table 1 compares results reported thus far for this formation from Fujian Province.

The presence of only a single polarity in this study and of Gilder et al. (1995) may indicate that these sites have been remagnetized. Between the two studies, a total of 35 samples from seven sites exhibit this behaviour. Until further data becomes available from this region, the two studies taken together (including the fact that both pass a fold test) should be considered as preliminary evidence of a primary Early Triassic direction for South China.

Results from this study are in good agreement with Gilder et al. (1995). In their interpretation of the results from the Xikou Formation, they prefer to place the sites in the northern hemisphere at the time that remanent magnetization was acquired (presuma-

Table 1
Mean declination and inclinations reported for the Xikou Formation in Fujian Province with $\alpha_{95}$s

| Study | Mean declination | Mean inclination | $\alpha_{95}$ |
|---|---|---|---|
| Zhai et al. (1991, 1994) | 335.5 | 4.9 | 7.9 |
| Gilder et al. (1995) (average of three methods) | 302.4 | −25.4 | 5.8 |
| This study | 302.3 | −20.5 | 8.4 |

bly the Early Triassic). This interpretation requires very significant counter-clockwise rotation (121°) and northward displacement (22°) vs. only 50° clockwise rotation and no significant displacement if the sites are in the southern hemisphere during the Early Triassic. Similar amounts of rotation are required to bring the results of this study into agreement with the Triassic paleopole from the Yangtze Block (58° clockwise if in the southern hemisphere and 122° counter-clockwise if in the northern hemisphere).

Gilder et al. (1995) have presented strong evidence supporting their interpretation; however, a separation of the Huanan and Yangtze Blocks during the Early Triassic cannot be ruled out. In fact, a separation of the two blocks is also supported by geologic evidence (Hsü et al., 1989, 1990) and requires significantly smaller amounts of rotation and translation. Until more reliable paleomagnetic results are obtained from south China, this question cannot be adequately resolved.

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